Capital Asset Pricing Model

Capital Asset Pricing Model (CAPM)

- The capital asset pricing model (CAPM) is used to determine a theoretically appropriate required rate of return of an asset, if that asset is to be added to an already well-diversified portfolio (market portfolio).
- The model takes into account the asset's sensitivity to market risk (also known as systematic risk or non-diversifiable risk), often represented by the quantity beta (β) in the financial industry, as well as the expected return of the market and the expected return of a theoretical risk-free asset.

The Market Portfolio

- In our last lecture, we constructed an efficient frontier, capital allocation lines, and the tangency portfolio using two stocks and the risk-free asset
- Suppose that we repeated the exercise with <u>all</u> securities, not just two. We would obtain a tangency portfolio, a capital allocation line (known as the security market line (SML))
- The market portfolio is the value-weighted portfolio of <u>all</u> securities (yes, <u>all</u> of them)...
- ...but in practice, we approximate the returns of the market portfolio with the returns of the S&P 500
- Investors <u>demand</u> a combination of the risk-free asset and the tangency portfolio; the market <u>supplies</u> the market portfolio and the risk-free asset
- In equilibrium, demand equals supply:

The tangency portfolio of all securities is the market portfolio

Systematic and Idiosyncratic Risks

Systematic Risk

- Systematic risk is risk to which <u>all</u> firms are exposed
- Apple examples: sales fall when unemployment rises, wages fall; conversely, sales rise when unemployment falls, wages rise
- Home Depot is exposed to the same risks

Idiosyncratic Risk

- Idiosyncratic risk is risk specific to the firm
- Apple examples: problems with Foxconn (an Apple supplier), Steve Jobs dies, the Justice Department figures out how to hack iPhones, etc.
- Home Depot is not exposed (directly) to these risks

The CAPM in a Nutshell

- In a nutshell, the CAPM says that

$$\overbrace{R_{\text{Stock}} - R_F}^{\text{Stock Return}} = \overbrace{\beta_{\text{Stock}} \times (R_{\text{Market}} - R_F)}^{\text{Systematic Risk}} + \overbrace{\epsilon_{\text{Stock}}}^{\text{Idiosyncratic Risk}}$$

  where ϵ_{Stock} is a random shock to the stock's return
- **Important:** $\text{E}[\epsilon_{\text{Stock}}] = 0$ and $\text{Cov}[R_{\text{Market}}, \epsilon_{\text{Stock}}] = 0$
- Taking expectations,

$$\overbrace{E[R_{\text{Stock}}] - R_F}^{\text{Stock Risk Premium}} = \beta_{\text{Stock}} \times \overbrace{(E[R_{\text{Market}}] - R_F)}^{\text{Equity Risk Premium}}$$

- Investors are <u>not</u> compensated for idiosyncratic risk: the stock's risk premium only depends on its beta (β_{Stock}) and the equity risk premium (a.k.a. market risk premium)

CAPM Assumptions

1. **Investor behavior**
   1.1 Investors are rational, mean-variance optimizers
   1.2 Their common planning horizon is a single period
   1.3 Investors all use identical input lists, and assumption often termed homogeneous expectations. Homogeneous expectations are consistent with the assumption that all relevant information is publicly available
2. **Market structure**
   2.1 All assets are publicly held and trade on public exchanges
   2.2 Investors can borrow or lend at a common risk-free rate, and they can take short positions on traded securities.
   2.3 No taxes.
   2.4 No transaction costs.

Beta

$$\overbrace{R_{\text{Stock}} - R_F}^{\text{Stock Return}} = \overbrace{\beta_{\text{Stock}} \times (R_{\text{Market}} - R_F)}^{\text{Systematic Risk}} + \overbrace{\epsilon_{\text{Stock}}}^{\text{Idiosyncratic Risk}}$$

- β_{Stock} (Beta) is the slope of the regression line obtained by regressing the stock's returns on the market's returns

$$\beta_{\text{Stock}} = \frac{\text{SD}[R_{\text{Stock}}]}{\text{SD}[R_{\text{Market}}]} \times \text{Corr}[R_{\text{Stock}}, R_{\text{Market}}] \tag{1}$$

- The beta of a security is the expected % change in its return given a 1% change in the return of the market portfolio
- It is the sensitivity of the stock's returns to systematic risk

CAPM Beta: Examples

Walmart Inc. (WMT) Add to watchlist

NYSE - NYSE Delayed Price. Currency in USD

96.90 +1.07 (+1.12%) At close: 4:00PM EDT

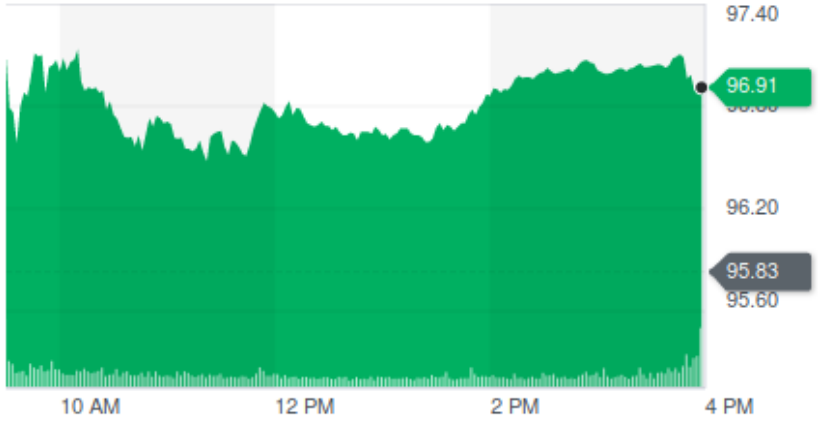
96.95 +0.05 (0.05%) After hours: 6:43PM EDT

Buy Sell

Summary | Chart | Conversations | Statistics | Profile | Financials | Options | Holders | Historical Data | Analysis | Sustainability

Previous Close	95.83	Market Cap	283.794B
Open	96.11	Beta	0.25
Bid	0.00 x 1100	PE Ratio (TTM)	55.43
Ask	0.00 x 3000	EPS (TTM)	1.75
Day's Range	96.09 - 97.19	Earnings Date	Nov 15, 2018
52 Week Range	77.50 - 109.98	Forward Dividend & Yield	2.08 (2.16%)
Volume	8,332,605	Ex-Dividend Date	2018-12-06
Avg. Volume	7,814,630	1y Target Est	104.00

1D 5D 1M 6M YTD 1Y 5Y Max Full screen



Walmart has a beta of **0.25**

CAPM Beta: Examples

Tiffany & Co. (TIF) Add to watchlist

NYSE - NYSE Delayed Price. Currency in USD

124.32 +1.45 (+1.18%) At close: 4:02PM EDT

124.09 -0.23 (-0.19%) After hours: 4:26PM EDT

Buy Sell

Summary | Chart | Conversations | Statistics | Profile | Financials | Options | Holders | Historical Data | Analysis | Sustainability

Previous Close	122.87	Market Cap	15.218B
Open	123.86	Beta	1.47
Bid	123.40 x 800	PE Ratio (TTM)	34.46
Ask	124.76 x 1400	EPS (TTM)	3.61
Day's Range	123.29 - 124.80	Earnings Date	Nov 27, 2018 - Dec 3, 2018
52 Week Range	86.15 - 141.64	Forward Dividend & Yield	2.20 (1.78%)
Volume	1,498,549	Ex-Dividend Date	2018-09-19
Avg. Volume	1,598,087	1y Target Est	140.50

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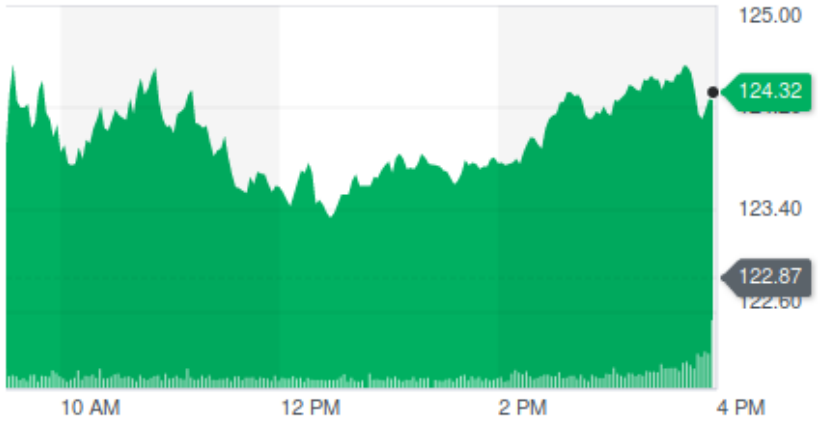

Tiffany's has a beta of **1.47**

Volaility Decomposition

- The stock's volatility is $\mathrm{SD}[R_{\mathrm{Stock}}]$
- The volatility of a porfolio with the same systematic risk as that of the stock is $\beta_{\mathrm{Stock}}\mathrm{SD}[R_{\mathrm{Market}}]$
- We say that a fraction

$$\frac{\beta_{\mathrm{Stock}}\mathrm{SD}[R_{\mathrm{Market}}]}{\mathrm{SD}[R_{\mathrm{Stock}}]} \tag{2}$$

  of the stock's volatility is systematic; we say that the remaining fraction is idiosyncratic

Example

Consider the following data for Cisco (CSCO) and the market

$$\begin{array}{ll} E[R_{CSCO}] = 0.9961\% & SD[R_{CSCO}] = 7.09\% \\ E[R_{MKT}] = 1.0773\% & SD[R_{MKT}] = 3.7568\% \\ Corr[R_{CSCO}, R_{MKT}] = 0.6339 & R_F = 0.0418\% \end{array}$$

1. What are Cisco's and the market's Sharpe ratios? .1346, .2756
2. What is Cisco's beta? 1.1963
3. What is the market's risk premium? 1.0355%
4. What is Cisco's risk premium? 0.9543%
5. What fraction of Cisco's volatility is systematic? .6339
6. What fraction is idiosyncratic? .3661

Example

Consider the following data for Caterpillar (CAT) and the market

$$\begin{array}{ll} \mathrm{E}[R_{\mathrm{CAT}}] = 1.3424\% & \mathrm{SD}[R_{\mathrm{CAT}}] = 8.1365\% \\ \mathrm{E}[R_{\mathrm{MKT}}] = 1.0773\% & \mathrm{SD}[R_{\mathrm{MKT}}] = 3.7568\% \\ \mathrm{Corr}[R_{\mathrm{CAT}}, R_{\mathrm{MKT}}] = 0.7279 & R_F = 0.0418\% \end{array}$$

1. What are Caterpillar's and the market's Sharpe ratios?
2. What is Caterpillar's beta?
3. What is the market's risk premium?
4. What is Caterpillar's risk premium?
5. What fraction of Caterpillar's volatility is systematic?
6. What fraction is idiosyncratic?

Alpha

- The difference between the stock's actual risk premium and the risk premium of a portfolio with the same systematic risk is the alpha:

$$\alpha_{\text{Stock}} = \underbrace{(\mathrm{E}[R_{\text{Stock}}] - R_F)}_{\text{True Value}} - \underbrace{\beta_{\text{Stock}} \times (\mathrm{E}[R_{\text{Market}}] - R_F)}_{\text{CAPM}} \quad (3)$$

- While one might observe a non-zero alpha in the short-term, it will likely become zero in the long-term

Example

1. What is Cisco's alpha? -0.2845%
2. What about Caterpillar?

The Efficient Market Hypothesis

The Efficient Market Hypothesis (EMH)

- The Efficient Market Hypothesis (EMH) states that prices of securities fully reflect available information about securities
- It highlights the difficulties associated with earning abnormal returns using either *fundamental analysis* (predicting returns based on accounting or economic data) or *technical analysis* (predicting returns based on the past history of returns).

The Random Walk

- A stock price follows a *random walk* returns are unpredictable.
- Suppose that Apple traded for 266 USD at 12:00 and everyone knew that at the end of the day, it would trade for 270 USD (a risk-free return of 1.5% in just one afternoon!)
- Everyone would start buying. The price would rise to 267 (a return of 1.13%), to 268 (.75%), and so on until at 12:01, when the price would be 270 (0%).
- Put differently, all information (such as the fact that the price will be 270 USD by the end of day) is immediately incorporated into the price
- The random walk explains why the stock market does well on days when bad news comes out (e.g. the market expected that unemployment would rise by 1%, traded stocks at prices reflecting that expectation, and then when unemployment rose by .95%, stock prices rose).

Forms of the EMH

There are three forms of the hypothesis, depending on how much information it assumes investors have

1. **Weak Form:** the assertion that stock prices already reflect all information contained in the history of past trading
2. **Semi-Strong Form:** the assertion that stock prices already reflect all publicly available information
3. **Strong Form:** the assertion that stock prices already reflect all relevant information including inside information

Technical Analysis

- Technical analysis involves researching recurrent and predictable stock price patterns
- It claims that even if prices on average incorporate all publicly information, they deviate in the short-term in predictable ways
- It's less about the long-term value of the security and more about its short-term supply and demand
- Technical analysts typically attribute these patterns to market psychology.
- History note: technical analysis has its roots in Charles Dow's "Dow theory" (Dow, of the Dow Jones Industrial Average and the Wall Street Journal).

Technical Analysis

- Common concepts in technical analysis include *support* (a price below which a stock's price doesn't typically fall) and *resistence* (a price above which a stock's price doesn't typically rise).
- It follows that when a stock's price nears its support, one should buy (it can only go up) and when the price nears its resistence, one should sell (it can only go down).
- **Important Caveat:** one factor identified in research is momentum: specifically the return of a portfolio that goes long stocks that have down well and short stocks that have done poorly. While this certainly has the flavor of technical analysis, the trading costs of such a strategy are high.

Support and Resistence

Consider a stock valued at 10 USD per share. Suppose that the chart shows a support of 9 USD and a resistence of 11 USD and suppose that because of a dividend increase, it is now valued at 15 USD per share. If other traders believe in a resistence level, then as soon as the price nears 11 USD, they will be willing to sell to you for 11 USD. If you buy at 11 USD, then you make a profit of 4 USD, as the present value of the cash flows is 15 USD. In order to realize this profit, the market must either give-up the resistence level or you have to hold the stock long-term and obtain the dividends yourself.

Fundamental Analysis

- Fundamental analysis involves research on the determinants of stock value, such as earnings and dividends prospects, expectations for future interest rates, and risk of the firm
- An extreme form of fundamental "analysis" involves collecting inside information (e.g. someone on the board tells you on the phone that he will announce a special dividend tomorrow or someone at the Fed tells you that the Fed will lower rates next week). This behavior is typically illegal.

Fundamental Analysis

Consider our example with support and resistence, but suppose now that the market did not believe in a support or resistence. Fundamental analysis would involve predicting the change in dividend, buying the stock when it traded for 10 USD and selling when it adjusted to 15 USD.

Technical and Fundamental Analyses: A Summary

- Technical analysis can easily be automated and traded *against*. One need only stock price data and a few milliseconds of computer time to identify trading opportunities and submit trade orders.
- Example algorithm: if everyone sells at an arbitrary resistence level, then submit a limit buy order at that level; if everyone buys at an arbitrary support level, then submit a limit sell order at that level. You will make money on average.
- Fundamental analysis, in constrast, requires much more work, and the people who do it are compensated for their labor in the form of alpha.

The Arbitrage Pricing Theory

The Arbitrage Pricing Theory

- These examples show that if $\alpha \neq 0$, then one can costlessly construct a portfolio that pays $|\alpha|$ on average.
- Suppose, for concreteness, that $\alpha > 0$ (as in the Walmart example). Then arbitrageurs start buying the stock.
- According to the laws of supply and demand, the price will rise (the return will fall) until $\alpha = 0$.
- The APT claims that $\alpha = 0$ on average:

$$\mathrm{E}[r_S - r_F] = \beta \mathrm{E}[r_M - r_F] \tag{4}$$

  which is exactly what the CAPM claims.
- The APT tells us *why* the CAPM should hold and *how* to profit when it breaks.
- Note that these trades involve idiosyncratic risk (ϵ) but <u>not</u> systematic risk (r_M)

Multifactor Models

- A multifactor model decomposes a security's return into *factors* $f_1, f_2, \ldots, f_N$:

$$r_S - r_F = \alpha + \beta_1 f_1 + \beta_2 f_2 + \cdots + \beta_N f_N + \epsilon \tag{5}$$

- The CAPM is a 1-factor model with $f_1 = r_M - r_F$.
- There are multifactor models for stocks (which we will explore in a moment), bonds, currencies, housing, and other securities.
- In currencies, for example, the two most popular popular factors are "value" and "carry", each corresponding to the return on a certain portfolio of currencies.
- Again, the APT claims that $\alpha = 0$ on average:

$$\mathrm{E}[r_S - r_F] = \beta_1 \mathrm{E}[f_1] + \beta_2 \mathrm{E}[f_2] + \cdots + \beta_N \mathrm{E}[f_N] \tag{6}$$

Application: Fama-French 3-Factor Model

- The Fama-French model is a multifactor model for stocks.
- It has factors $f_1 = r_{\text{MKT}} - r_F$, $f_2 = r_{\text{SMB}}$, and $f_3 = r_{\text{HML}}$.
- SMB is a portfolio that goes long (S)mall stocks and short (or (M)inus) (B)ig stocks (as measured by market cap).
- Smaller firms outperform bigger firms (3.06% 1927-2019).
- HML is a portfolio that goes long (H)igh book-to-market stocks and short (or (M)inus) (L)ow book-to-market stocks.
- Two definitions: *value stocks* are stocks with a (H)igh book-to-market ratio, while *growth stocks* are stocks with a (L)ow book-to-market ratio.
- Value stocks outperform growth stocks (4.48% 1927-2019).
- The idea is that smaller stocks and stocks with higher book-to-market are riskier and therefore offer higher returns
- `https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html`